Nonoverlapped Sampling for Hyperspectral Imagery: Performance Evaluation and a Cotraining-Based Classification Strategy

Xianghai Cao, *Member, IEEE*, Zuji Liu, Xiangxiang Li, Qian Xiao, Jie Feng, *Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— For hyperspectral imagery (HSI) classification, most of the studies focus on how to improve the classification accuracy, while the influence of sampling strategy for classification performance attracts little attention. For now, random sampling (RS) is the most adopted strategy. That is, for a hyperspectral image, a certain number of labeled samples are randomly selected as the training set, and the remaining labeled samples are taken as the test set. However, the RS strategy will produce over optimistic results when used for performance evaluation because of the overlap between training set and test set. Though spectral-spatial classification methods benefit most from the RS strategy, the pixel-wise classification methods can also benefit from it because of the high spectral correlation between training and test samples. However, in practical applications, the RS strategy is not feasible. Because the training and test samples are often collected from different locations. In this situation, the correlation between training and test samples will decrease dramatically and the performance of HSI classification methods will be affected. In this article, a nonoverlapped sampling method is adopted to reduce the correlation between training and test samples and different classic classification methods are evaluated. Experimental results show that the classification performance of all methods drops a lot when nonoverlapped sampling strategy is adopted. After the analysis of some important factors for HSI classification, we also propose a cotraining-based classification method to relief the influence of sampling strategy and obtains much better performance compared with those classic spectral-spatial classification methods.**

***Index Terms*— Hyperspectral imagery (HSI), performance evaluation, sampling strategy, spectral correlation.**

I. Introduction

WITH the advance of remote sensing technology, the spectral and spatial resolution of hyperspectral sensors are getting higher and higher. Therefore, the captured hyperspectral images can provide abundant information for many practical applications, such as precision agriculture [1], resource exploration [2] and urban planning [3]. Hyperspectral imagery (HSI) classification is one of the most used the method in the above applications, because it can provide accurate label information for each pixel. Based on whether the spatial context information is used, the HSI classification methods can be classified into two categories: pixel-wise method and spectral-spatial fusion method.

The pixel-wise method adopts spectral feature as the input of the classifier. Bayesian models [4], feature extraction and feature reduction techniques [5], random forest [6], neural networks [7] and kernel methods [8] have been investigated for the classification of HSI. Among these methods, the support vector machines (SVM) classifier shows a good performance with relatively few training samples to obtain good classification accuracies. However, the pixel-wise method only exploits the spectral information for classification, their accuracy is far from satisfactory.

For HSI, an important characteristic is local smoothness, that is, the neighbor pixels often share the same label with high probability. With the local smoothness, many spectral-spatial methods are proposed to improve the HSI classification accuracy where the pixel and its neighbor pixels are exploited together to extract discriminating features. There are diverse techniques for spectral-spatial classification. Some variations based on SVM were proposed. These methods combine spectral and textural information based on kernel matrix, such as composite kernels [9], graph kernels [10], and multiple kernels [11]. Markov random fields (MRFs) [12] use local spatial neighborhood to construct spatial energy term and pixel-wise classification probability to construct classification term [13], [14], the energy function is minimized to find the smoothed classification result. Spatial filtering is another simple method to exploit the local smoothness, which is used to reduce salt-pepper noise and improve classification accuracy effectively based on the assumption that there is usually a strong correlation among neighboring pixels [15]. In [16], edge-preserving filtering was used to process the probability maps obtained by the SVM classifier and the classification result is determined by finding the maximum probability. Many spectral-spatial methods exploit the spatial context information to extract meaningful features. For example, the classic extended morphological profile (EMP) [17] is extracted to isolate bright and dark structures in images. A multilevel, multiattribute approach based on morphological

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attribute filters named as extended morphological attribute profile (EMAP) [18] was also proposed to characterize the spatial information. The feature descriptors, e.g., 3-D Gabor wavelet [19] and local binary patterns [20] calculate features by using spatial neighbor pixels. Besides, some methods extract different spectral-spatial features and then fuse them to exploit the complement information [21], [22]. The sparse representation model selects some training samples to form the dictionary and represent test samples [23], [24]. These methods incorporate the spatial smoothness of the neighbor pixels directly in the sparse recovery problem. Active learning techniques can reduce the cost of acquiring a large number of labeled training samples [25], [26]. The purpose of active learning is to gradually improve the performance of the model through sampling, and obtains the training set with most information. The classifier with active learning is trained on a small group of carefully selected samples and can be as good as the classifier on a large number of randomly selected samples.

Although the classification accuracy of HSI has been improved a lot with the spectral-spatial combination, the drawback is that their constructed features are handcrafted and show limited generalization ability. In the last few years, deep learning-based approaches have been used for HSI classification motivated by the success in the fields of image, video, and speech recognition [27]–[29]. The convolutional neural networks (CNNs) [30] are multilevel deep learning models that use an end-to-end training framework. In [31]–[34], CNNs exploit 1-D, 2-D, and 3-D features to perform HSI classification. In [35], the 2-D CNN and 3-D CNN are combined to improve the classification performance. In [36], an HSI classification method based on information measure and CNN is proposed. Other networks such as spectral-spatial long short term memory (LSTM) networks [37] and recurrent neural networks (RNNs) [38] are also introduced for HSI classification.

The spatial-spectral methods have greatly improved the accuracy of hyperspectral datasets. However, because of the high cost of collecting ground truth information of HSI, almost all classification methods select training and test samples from the same image. The most used sampling strategy at present is random sampling (RS). That is, part of labeled samples are randomly selected from an HSI as the training set, and the rest labeled samples are taken as the test set. For spectral-spatial methods, because of the use of spatial context information, the training and test set are highly overlapped with each other. This can be observed from Fig. 1. If a 5×5 neighborhood is adopted to extract spectral-spatial features, most test samples will be involved during the feature extraction of the training set. Though the extracted features present excellent classification performance, it violates the basic rule of supervised learning that training samples should be independent with test samples. This means that the improvement of the classification accuracy of the spatial-spectral methods is partly due to the explicit or implicit use of test sample information during the training process. However, in practical applications, the training samples and test samples may be collected from different locations, different times, or even with different sensors. This means the nonoverlapped sampling strategy is a reasonable choice. In this situation, the high correlation between training and test samples doesn't exist anymore. So, with the RS strategy, the evaluation results cannot present the actual classification performance when these spectral-spatial methods are adopted for practical applications. In fact, even without using the spatial context information, because the training samples and test samples are spatial adjacent to each other, the pixel-wise methods still produce optimistic results because of the high spectral similarity between training samples and test samples.

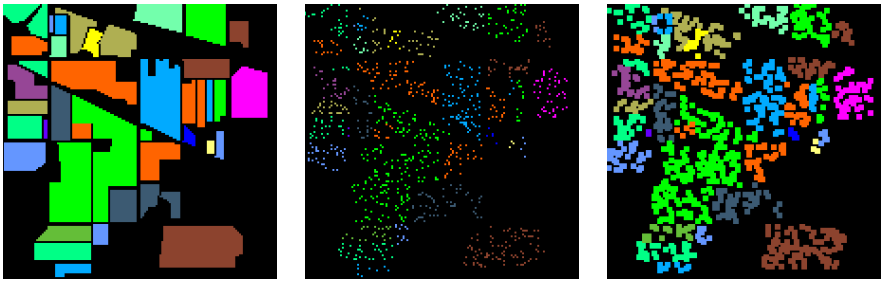
Fried *et al.* [39] may be the first to point out that the training samples and test samples with high correlation will lead to optimistic results. Liang *et al.* [40] explained in detail that RS would lead to overlap between training samples and test samples, and proposed a new nonoverlapped sampling method, named controlled random sampling (CRS), which can reduce the overlap a lot. In addition, to ensure that the training set and test set disjoint, patch sampling [41], and clustering sampling [42] were also proposed. Though some researchers have noticed this problem and proposed some nonoverlapped sampling strategies to reduce the overlap between training and test samples, a comprehensive performance evaluation of HSI classification methods based on nonoverlapped sampling has not been presented. In this article, in order to evaluate the actual performance when HSI classification methods are used in practical situations, the CRS sampling method [40] is adopted which can minimize the overlap between the training and test set. To give a fair and authentic result, the three most widely used datasets and seven representative HSI classification methods are used and four important factors for HSI classification are presented for performance analysis. After the deep analysis of the experimental results, a new classification method is proposed to solve the nonoverlapped HSI classification problem. The novelty of this article is as follows: 1) A comprehensive performance evaluation of some representative and different HSI classification methods with RS and nonoverlapped sampling strategy is provided. 2) Four important factors for HSI classification are presented to explain the performance difference between RS and nonoverlapped sampling-based methods. 3) Based on the factor analysis, a cotraining method is proposed to exploit the spatial adjacency to improve the classification accuracy of nonoverlapped sampling-based method.

The remainder of this article is organized as follows: Section II describes the nonoverlapped sampling strategy and several classic classification approaches adopted in the experiments. Section III presents the performance comparison under RS and nonoverlapped sampling. Section IV proposes a cotraining-based nonoverlapped HSI classification method and presents its classification performance. Section V concludes this article.

II. Related Work

In this section, the CRS method is briefly described. Then, several representative HSI classification methods are introduced for performance evaluation.

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(a) (b) (c)

Fig. 1. RS strategy with Indian pines. (a) Ground truth. (b) RS. (c) RS with 5×5 neighbor size.

A. Controlled Random Sampling

RS is not applicable in practical applications. The main problem is that the training samples and test samples are randomly selected from the same image, resulting in a high overlap and correlation between training and test samples. In order to evaluate the actual performance of the HSI classification methods, a nonoverlapped sampling strategy should be adopted. In an ideal setting, the training and test samples should be selected from different images. However, these hyperspectral datasets which share the same land cover types and have accurate label information are not available. So, we still select the training and test samples from the same image and introduce a nonoverlapped sampling strategy to minimize the overlap between test set and training set. Specifically, the CRS is adopted in this article [40]. For HSI, each category usually distributes in several partitions on the image. With CRS, seeds are randomly selected in each image patch to ensure the dispersion globally of the training set. Then, the seed points adopt the idea of region growing to increase the training set gradually. In this way, the training set is gathered tightly rather than scattered randomly in the whole image. The CRS method greatly reduces the spatial correlation between the test set and training set and makes the evaluation more believable. The visual comparison between RS and nonoverlapped sampling can be found in Figs. 2–4.

B. Classification Methods

In this part, seven representative and diverse classification methods used for performance evaluation are introduced briefly.

1) SVM: The SVM classifier finds the optimal separation hyperplane in the Hilbert space so as to maximize the positive and negative sample spacing of the training set. SVM is a supervised learning algorithm used to solve binary classification problems. In order to solve the multiclass classification problem, the dominant approach for doing so is to reduce the single multiclass problem into multiple binary classification problems. Common methods for such reduction include: Building binary classifiers that distinguish between one of the labels and the rest (one-versus-all) or between every pair of classes (one-versus-one). After the introduction of kernel methods, SVM can be used to process nonlinear problem effectively. Studies have shown that SVM is very suitable for hyperspectral image classification and is less affected by the Hughes phenomenon.

2) SVM-CK: SVM-based kernel methods can effectively deal with large input spaces with relatively few labeled training samples, so it is suitable to solve the problem of hyperspectral data classification. However, the kernel method can be further improved by including contextual information. Camps-Valls *et al.* [43] proposed composite kernel method to fuse the spectral and spatial information. The theoretical support is that two (or more) Hilbert Spaces can be directly summed and merged into a larger Hilbert space. This method enhances the classification accuracy as compared to traditional approaches that take into account the spectral information only. Besides, the composite kernel method is flexible to balance between the spatial and spectral information in the classifier.

3) MRF: For hyperspectral image, MRF can model the spatial neighborhood of pixels as a random process of spatial distribution. In order to improve classification accuracy, many methods use MRF to refine the pixel-wise classification results. In [44], the work combined SVM and MRF into a unified framework, and it first performed classification in the spectral domain with SVM classifier and then fed the resulted posterior probabilities to the MRF spatial analysis stage. The purpose of MRF is to assign each pixel to the class with the maximum posterior probability. This can be achieved by minimizing an energy function. The energy function includes spatial energy term and spectral energy term, where spectral energy term is calculated from pixel-wise classification probabilities and spatial energy term is constructed based on the local neighborhood. A detailed description can be found in [44].

4) EMP: Morphological profile (MP) consists of a series of opening profiles and closing profiles. The opening and closing operations remove the image structure that cannot contain the structuring elements. To include spectral information in MP, several images containing partial spectral information are usually extracted, and then MP is constructed with each individual image to obtain the EMP. In hyperspectral image classification, [17] extracted several principal components (PCs) which contain a certain amount of spectral information, and performed opening and closing operations on each PC to build MP. The EMP can be constructed by stacking MP vectors of all PCs, which can be used as the classification feature.

5) EPF: Edge-preserving filter can smooth the local similar region of an image while keeping most edge information.

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In [16], the edge-preserving filter is used to introduce the spatial information for HSI classification. First, the SVM classifier is used to obtain the pixel-wise classification result which is then used to form multiple probability maps. The probability map is smoothed with the edge-preserving filter to introduce the spatial context and improve the classification performance. Finally, each pixel is classified by finding the maximum probability. This method is simple and effective for HSI classification.

6) JSRC: Sparse representation is introduced into hyperspectral data processing based on the observation that a HSI pixel can be sparsely represented by a linear combination of a few training samples from a given dictionary. In order to improve the classification performance, the contextual information is also incorporated into the sparse recovery optimization problem. A joint sparsity model (JSRC) [23] was proposed that the test pixels in a small neighborhood are simultaneously represented as a linear combination of training samples, with different weighted coefficients for each sample. By solving the sparse constraint optimization problem, the sparse vector is recovered and the label of each test pixel is determined as the one with the minimal residual.

7) CAE: Stacked autoencoders are widely used for feature extraction. In [45], a two stream architecture is proposed for HSI classification, one stream employs the stacked denoising autoencoder to encode the spectral values of each input pixel, and the other stream takes as input the corresponding image patch and deep CNNs are employed to process the image patch. In the fusion scheme, the prediction probabilities from two streams are fused by adaptive class-specific weights. Besides, a regularizer is added to the loss function to alleviate the overfitting.

8) CRNN: By considering the spectral signature as a sequence, RNNs have been successfully used to learn discriminative features from hyperspectral images. However, most of these methods only input the whole spectral bands into RNNs directly, which may not fully explore the redundant and complementary information of HSI. In [46], a cascaded RNN (CRNN) model is proposed for HSI classification. Two RNN layers are used to eliminate redundant information and learn the complementary information between spectral bands, respectively. By incorporating some convolutional layers, the spatial information is used to improve the classification performance of HSI.

For HSI classification, four factors contribute to the classification performance. 1) Spectral feature: Spectral information is the foundation of HSI processing and determines the final classification performance. 2) Spectral-spatial fusion: This strategy combines spectral feature and spatial context to increase the distance of the different class and decrease the distance between the same class, which is the most used way to improve the classification performance of HSI. 3) Spatial adjacency: With RS, the training samples and test samples are randomly selected from the same image. This means that training and test samples are spatial adjacent to each other and have high spectral similarity. 4) Samples overlap: For spectral-spatial combined classification model, if the RS is adopted, during training stage, many test samples will be used because of the introduction of spatial context, and vice versa. In the following experiments, we will analyze the classification results based on these four factors.

III. Performance Evaluation With RS and Nonoverlapped Sampling

A. Datasets

The classification methods are compared with three widely used datasets: the Indian pines scene, the Salinas scene, and the Kennedy Space Center (KSC) scene. The details of these datasets are as follows.

The Indian Pines scene was gathered by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor. The image size is 145×145 and includes 224 spectral bands with wavelength ranges from 400 to 2500 nm. The Indian Pines scene contains two-thirds agriculture, and one-third forest or other natural perennial vegetation. The scene includes sixteen classes and the number of bands is reduced to 200 by removing bands covering the region of water absorption. The composite color image, reference map, sampling map, and legend are presented in Fig. 2.

The size of the Salinas scene is 512×217. It contains 224 bands and the spatial resolution is 3.7 m. The number of bands is corrected to 204 by omitting 20 noise bands (108–112, 154–167, and 224). The ground truth consists of 16 land cover classes. Fig. 3(a) is a false-color composite of the scene and 3(b) is the ground truth. Fig. 3(c) and (d) are the training set under two sampling strategies. Fig. 3(e) shows the legend of 16 different land cover types.

The KSC acquired from an altitude of approximately 20 km, has a spatial resolution of 18 m. After removing water absorption and low SNR bands, 176 bands were used for analysis. For classification purposes, 13 classes representing the various land cover types that occur in this environment were defined for the site. Fig. 4 provides the visual demonstration of this dataset.

B. Evaluation Indexes

In order to provide a fair performance evaluation, overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used. The OA is defined as the number of correctly predicted items/total of the item to predict, which is the most used indicator for HSI classification. The AA is defined as the average of each accuracy per class. Kappa coefficient is a statistic that is used to measure interrater reliability for qualitative items. It is generally thought to be a more robust measure than simple percent agreement calculation, as kappa takes into account the possibility of the agreement occurring by chance. The classification accuracy of each category is also provided for comparison.

C. Influence of Different Sampling Strategy for HSI Classification

Except for the JSRC, convolutional autoencoder (CAE), and CRNN, all methods use the SVM as a classifier. For JSRC, the reconstruction error is used to determine the

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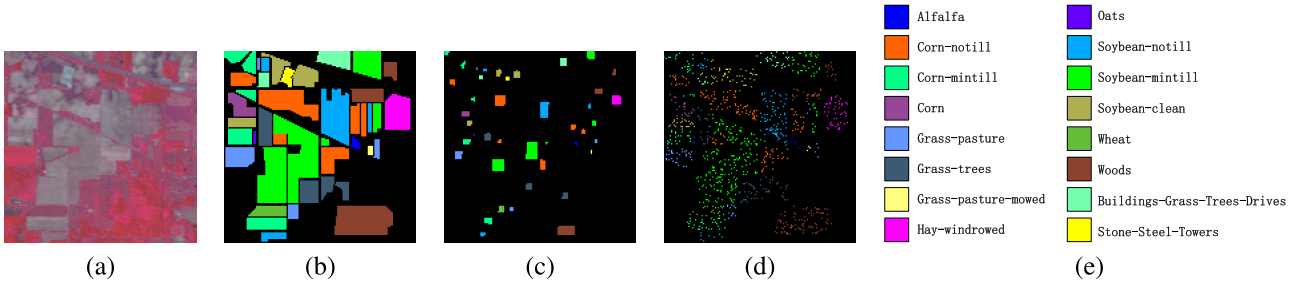


Fig. 2. Indian pines. (a) Three-band color composite image. (b) Ground truth. (c) Training set with CRS. (d) Training set with RS. (e) Land cover types.

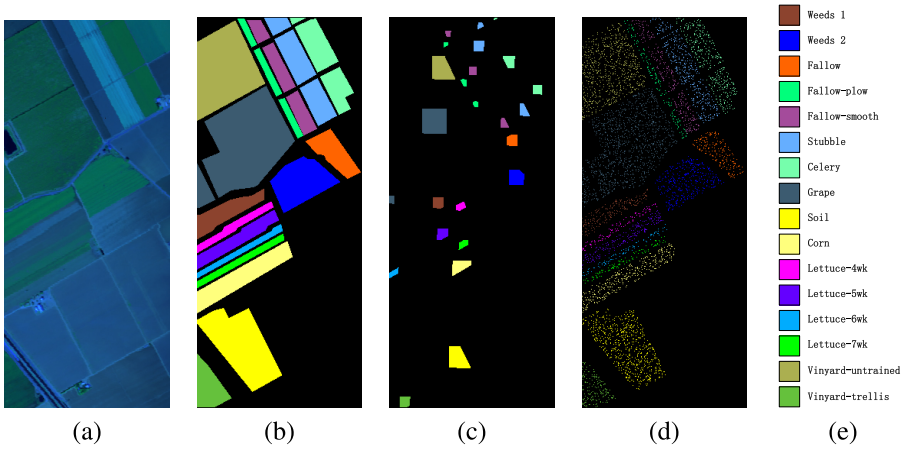


Fig. 3. Salinas. (a) Three-band color composite image. (b) Ground truth. (c) Training set with nonoverlapped sampling. (d) Training set with RS. (e) Land cover types.

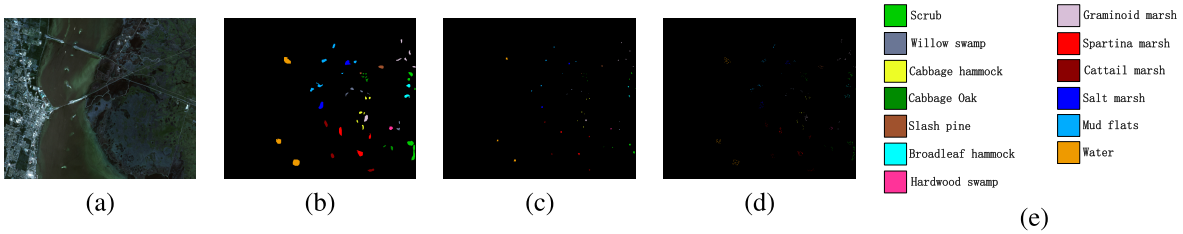


Fig. 4. KSC. (a) Three-band color composite image. (b) Ground truth. (c) Training set with nonoverlapped sampling. (d) Training set with RS. (e) Land cover types.

pixel label. For CAE and CRNN, softmax is used for classification. The SVM is implemented with the library for SVM (LIBSVM) library [47], and the Gaussian kernel is adopted. The SVM parameters are optimized by fivefold cross-validation. For SVM with composite kernels (SVM-CK) [43] method, the window size of all datasets are 9×9, and both the weights of the spatial kernel and spectral kernel are 0.5. For MRF [44], eight neighborhoods are selected to introduce the spatial regularization, and the context weight parameters of all three datasets are set as 1. The parameters of EMP and edge-preserved filtering (EPF) are the default values presented in [17] and [16]. For JSRC, the sparsity levels for all datasets are chosen as 3 and the window scale is selected as 9×9. For CAE and CRNN, the patch size is also 9×9. The CRS method is adopted as the nonoverlapped sampling strategy in the following experiments [40].

For the Indian Pines scene, 10% labeled pixels are used as the training set, and the rest are used for test. Experimental results are listed in Table I. With RS, the OA of pixel-wise SVM classifier is 80.02%. All spectral-spatial methods have obvious performance improvement by introducing spatial context information. With the spectral-spatial information fusion, the OA of MRF is 97.25%, which is 17.23% higher than that of the pixel-wise SVM classifier. For the worst performed EMP, the OA still improves from 80.02% to 88.02%. From this result, the conclusion can be easily obtained: spectral-spatial fusion can improve the HSI classification performance a lot. However, when nonoverlapped sampling strategy is adopted, all algorithms have a huge performance drop, even for the pixel-wise SVM classifier. The OA of SVM is 67.52%, which is 12.50% lower than the RS-based SVM classifier. In fact, because the input feature of pixel-wise SVM is only the spectral feature, so the training and test samples have no overlap with each other. The reason is that with RS, the training and test samples are spatial adjacent to each other and their spectral features are highly correlated. With

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nonoverlapped sampling, the spatial adjacency does not exist anymore and the spectral correlation between training and test samples also decreases a lot. Different from RS strategy, two spectral-spatial methods: SVM-CK and CRNN do not improve the classification accuracy with the introduction of spatial information. This is because training and test samples are not spatially adjacent to each other and the similarity of spectral features is low. In this situation, SVM-CK and CRNN cannot exploit the spatial information effectively any longer. The other five spectral-spatial methods: MRF, EMP, JSRC, EPF, and CAE still achieve some performance improvement with spatial context. This means that spectral-spatial fusion is still an effective way to improve the HSI classification accuracy for CRS sampling strategy. Take the MRF for example, when CRS is adopted, the MRF obtains the OA of 80.29%, which is 12.77% higher than that of SVM.

Similar to the Indian Pines scene, the Salinas scene has a large continuous area for different land cover types which can be observed from Fig. 3. For this data, only 5% labeled samples are used for training because these data contain much more samples than Indian Pines. The classification results are shown in Table II. For the widely used RS strategy, the OA of SVM is 92.50% and all spectral-spatial methods have better performance when compared with it. With the spectral-spatial fusion, the MRF almost correctly classifies all test samples with an OA of 99.49%. When the CRS is used, the OA of SVM reduces from 92.50% to 82.20%. Similar to the Indian Pines, the SVM-CK and CRNN still cannot improve the classification performance effectively by introducing the spatial context. All other spectral-spatial methods can provide performance promotion more or less. So, we can say that spectral-spatial fusion is still an effective way to promote the classification performance of HSI when CRS is adopted for sampling. Though RS can provide much better performance compared with CRS, it is not applicable in practical applications.

For the KSC dataset, 10% labeled samples are used for training and the rest 90% labeled samples are used for the test. Different from Indian Pines and Salinas, the KSC scene consists of many small and scattered areas. So, when RS is adopted, the correlation between training samples and test samples is much lower compared with Indian Pines and Salinas. As a result, the performance drop is also much lower when RS strategy is replaced with nonoverlapped sampling strategy. Take the SVM classifier, for example, the OA is reduced from 91.23% to 86.06%. For both RS and nonoverlapped sampling, except the EMP method, all other methods can improve the classification by exploiting the spatial information.

Let's use the SVM and MRF as an example to make further analysis to find the influence of spectral feature, spectral-spatial fusion, spatial adjacency, and sample overlap. For a clear presentation, their classification results based on Salinas are shown in Fig. 5. For CRS-SVM, the OA is 82.20%, which is obtained just with the spectral feature. By introducing the spatial information, the CRS-MRF obtained 85.45% accuracy, which is 3.25% higher than that of CRS-SVM. This improvement is benefited from spectral-spatial fusion. For RS-SVM, the OA is 92.50%. Though there is no overlap between training and test samples, the training

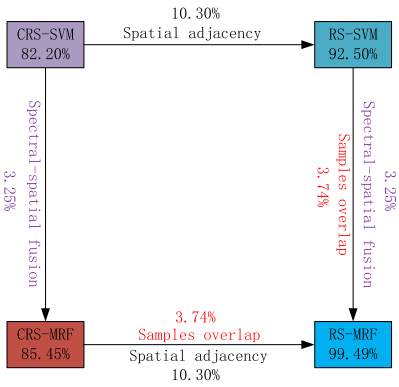


Fig. 5. Classification contribution of different factors.

and test samples are spatial adjacent to each other, so their spectral feature is highly similar. That is, the spatial adjacency brings accuracy promotion of 10.30%. Compared with RS-SVM, the OA of RS-MRF is 99.49%, which is the result of sample overlap and spectral-spatial fusion. We have known that spectral-spatial fusion improves the OA by 3.25%, so the accuracy improvement caused by sample overlap should be 3.74%. From the above analysis we can conclude that spatial adjacency and spectral-spatial fusion are the key to improve the HSI classification accuracy. For the four factors, the samples overlap is not available and should not be used for HSI classification. The spectral-spatial fusion is still a useful strategy for HSI classification. However, the spatial adjacent cannot be used directly with nonoverlapped sampling, so we need to find new strategy to exploit this important classification factor.

D. Distribution Comparison of Nonoverlapped Sampling and RS

In fact, with different sampling strategy, the feature space shows different distribution characteristics. In order to clarify this clearly, the Salinas data is adopted. The EPF feature is adopted and the widely used t-stochastic neighborhood embedding technique (SNE) [48] method is used to demonstrate the distribution of training and test samples with different sampling strategy. The experimental results are shown in Fig. 6. In this figure, different classes are shown with different colors and the training samples are highlighted with circles. We can find that for the RS strategy, the training samples are evenly distributed in the whole sample space. However, with nonoverlapped sampling strategy, the training samples often concentrate in some local areas. With Fig. 6, it is easy to understand that the performance degradation when nonoverlapped sampling is adopted for HSI classification.

E. Is Nonoverlapped Classification Only a Small Sample Problem?

Nonoverlapped sampling reduces the correlation between training samples and test samples. If the number of the training samples is reduced, then the correlation between training and test samples will be reduced also. So, whether the classification performance of nonoverlapped sampling and

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TABLE I

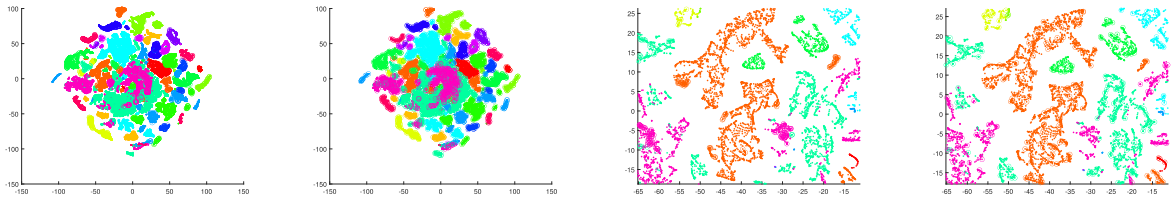
INDIAN PINES: THE CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS WITH 10% TRAINING SAMPLES

	Non-overlapped Sampling								Random Sampling							
Categories	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN
Alfalfa	43.75	75.10	20.00	59.46	75.64	42.19	73.88	49.36	58.60	99.07	70.00	91.66	88.03	98.08	95.12	95.21
Corn-notill	55.47	47.17	68.06	59.85	68.55	61.45	64.85	59.09	74.20	92.61	99.02	79.29	89.53	94.62	92.89	95.44
Corn-mintill	57.70	47.34	80.18	57.55	69.00	66.72	59.73	48.08	74.94	93.08	96.94	78.50	88.77	91.14	95.18	93.55
Corn	49.63	73.83	94.18	37.91	75.02	77.98	47.34	33.00	61.69	93.06	98.38	66.22	85.46	88.75	89.04	90.05
Grass-pasture	74.79	74.98	79.55	70.48	81.38	73.56	71.07	47.22	88.59	97.01	95.12	89.30	88.75	93.92	97.73	96.64
Grass-trees	84.44	85.87	95.80	94.09	86.58	91.54	81.27	80.01	88.90	97.90	97.76	96.10	93.37	97.76	97.61	97.19
Grass-pasture-mowed	65.87	56.68	27.88	67.86	63.50	76.95	68.57	45.93	89.44	93.70	30.00	85.83	78.12	95.21	99.57	93.44
Hay-windrowed	89.31	98.98	96.34	95.16	98.80	95.04	94.19	95.87	89.36	99.45	97.56	98.15	98.97	97.07	96.65	98.97
Oats	0.000	43.43	0.000	43.62	13.57	0.000	79.08	19.39	0.000	92.62	0.000	54.91	33.04	60.00	97.08	92.93
Soybean-notill	58.88	51.41	74.25	66.29	71.25	64.87	58.31	51.49	78.04	92.29	97.50	85.57	89.85	92.30	92.86	92.97
Soybean-mintill	71.55	70.96	86.75	84.25	78.54	80.00	69.74	64.11	77.51	95.85	96.46	91.37	95.17	92.10	95.26	95.39
Soybean-clean	49.73	52.41	63.88	50.44	75.79	55.06	46.42	40.37	68.30	93.85	94.41	76.16	90.18	89.10	91.69	91.74
Wheat	93.54	98.52	100.0	97.94	93.89	98.57	89.43	78.60	94.84	99.84	100.0	99.07	89.77	99.78	99.35	99.63
Woods	88.15	86.35	91.56	93.37	84.85	89.47	83.21	78.07	91.41	97.75	99.50	97.88	95.46	97.38	95.96	97.97
Buildings-Grass-Trees	56.73	75.79	84.86	79.72	81.47	69.40	62.05	57.32	68.74	95.67	93.51	90.47	95.52	89.06	90.78	92.01
Stone-Steel-Towers	96.47	97.67	57.90	98.33	69.13	98.40	87.49	83.54	98.84	98.90	98.56	98.12	66.60	97.83	95.96	91.26
OA	67.52	65.13	80.29	73.32	77.40	73.50	68.86	60.49	80.02	95.33	97.25	88.02	91.97	93.54	94.64	95.20
AA	64.75	71.03	70.07	72.27	74.16	71.32	71.04	58.22	75.21	95.79	85.29	86.16	90.84	92.13	95.17	94.65
Kappa	0.630	0.605	0.775	0.697	0.742	0.694	0.643	54.77	0.771	0.947	0.969	0.863	0.854	0.926	93.88	94.52

TABLE II

SALINAS: THE CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS WITH 5% TRAINING SAMPLES

	Non-overlapped Sampling								Random Sampling							
Categories	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN
Weeds 1	96.68	92.55	100.0	97.19	98.37	99.39	94.93	84.16	99.91	99.91	100.0	100.0	97.95	100.0	99.95	99.53
Weeds 2	98.44	98.53	99.53	99.24	98.51	98.97	97.95	91.33	99.60	99.78	99.97	99.85	100.0	99.75	99.93	99.59
Fallow	90.71	83.69	95.86	95.25	95.66	93.22	88.00	76.48	96.74	98.29	99.47	98.87	99.85	98.39	98.65	98.83
Fallow-plow	97.71	96.10	98.15	98.29	92.00	97.92	98.04	87.82	98.78	98.48	99.04	98.93	92.39	98.68	99.66	99.04
Fallow-smooth	88.10	78.34	89.56	88.58	72.39	88.87	84.12	90.10	98.89	99.00	99.48	99.34	96.93	99.16	99.62	99.46
Stubble	99.95	99.86	99.97	99.83	95.08	99.95	99.03	99.58	99.97	99.95	99.97	99.99	98.74	99.96	99.96	99.99
Celery	94.77	92.06	96.03	97.94	98.28	95.34	93.15	87.68	99.61	99.59	99.75	99.95	99.81	99.48	99.71	99.77
Grape	70.89	72.67	74.91	82.33	72.81	73.66	75.58	73.13	81.24	91.93	99.31	98.13	99.35	96.09	94.65	95.41
Soil	98.30	96.36	99.33	98.97	94.42	99.06	98.61	94.83	99.42	99.64	99.78	99.66	99.73	99.75	99.95	99.92
Corn	77.07	87.33	82.17	82.48	93.10	77.37	88.27	86.69	95.12	98.46	98.40	98.14	97.91	97.11	99.21	99.13
Lettuce-4wk	83.44	79.88	83.01	77.43	82.63	77.07	88.90	61.06	97.44	98.46	99.95	99.49	96.76	99.54	98.61	98.97
Lettuce-5wk	86.47	85.85	93.81	95.83	96.09	90.23	85.68	81.95	98.72	99.40	99.70	99.75	98.44	99.30	99.89	99.90
Lettuce-6wk	96.13	88.07	99.21	97.76	77.40	97.38	97.26	88.39	97.42	99.38	100.0	99.86	95.20	98.92	99.84	100.0
Lettuce-7wk	94.09	91.57	96.48	96.22	78.97	93.94	98.82	90.12	96.89	98.71	98.43	98.99	95.55	97.59	99.40	99.64
Vinyard-untrained	54.67	54.46	64.36	76.79	63.78	61.69	53.82	52.73	79.66	88.58	99.22	97.20	99.07	95.23	91.71	91.65
Vinyard-trellis	99.69	96.21	100.0	99.99	100.0	99.48	97.45	90.04	98.77	99.43	99.91	99.38	100.0	98.99	99.60	99.81
OA	82.20	81.58	85.45	88.68	84.18	83.93	83.03	79.00	92.50	96.34	99.49	98.90	98.75	98.01	97.54	97.64
AA	89.19	87.10	92.02	90.25	82.34	90.25	89.98	83.51	96.14	98.06	99.52	99.22	98.61	98.62	98.77	98.79
Kappa	0.802	0.795	0.838	0.874	0.881	0.821	0.812	76.69	0.916	0.959	0.994	0.988	0.980	0.978	97.26	97.38



(a) (b) (c) (d)

Fig. 6. Spatial distribution of Salinas. (a) Nonoverlapped sampling. (b) RS. (c) Local part of CRS. (d) Local part of RS.

RS-based methods will become the same when only very few training samples available? In this article, the Salinas scene is used to illustrate this problem. In this experiment, only 0.2% labeled samples are used for training. This means for most land cover types, only 4–6 pixels are used for training. The experimental results are shown in Table IV. We can see that though very few samples are used for training, the performance difference between RS and nonoverlapped sampling-based methods is still very large. Take the SVM for example, with nonoverlapped sampling, the OA is only 69.17%, however,

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TABLE III
KSC: THE CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS WITH 10% TRAINING SAMPLES

Categories	Non-overlapped Sampling								Random Sampling							
	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN
Scrub	90.76	96.31	98.03	91.94	86.58	95.23	94.28	96.32	93.16	99.22	98.54	91.26	98.77	97.01	99.02	99.18
Willow swamp	81.17	86.67	90.72	63.40	96.35	88.81	85.45	84.10	88.00	98.69	96.69	76.29	99.19	94.39	97.17	94.25
Cabbage hammock	86.05	92.13	98.52	76.41	91.17	97.25	66.09	78.91	89.45	96.57	99.79	79.55	98.06	96.70	84.47	95.53
Cabbage Oak	57.68	60.67	63.37	52.68	77.12	61.70	55.26	60.49	65.47	77.85	79.46	63.61	96.93	77.52	76.67	87.37
Slash pine	56.67	62.48	65.59	72.76	91.13	67.41	56.74	65.12	78.88	83.09	77.78	76.09	98.53	78.97	82.13	91.70
Broadleaf hammock	57.83	81.42	92.02	62.11	81.94	86.55	73.44	78.18	74.33	91.54	98.99	77.69	98.49	94.83	87.60	95.28
Hardwood swamp	75.63	92.38	86.67	51.80	97.23	90.06	80.53	75.99	79.85	98.37	97.46	69.02	97.91	88.22	92.66	92.25
Graminoid marsh	76.43	83.18	86.59	83.41	96.47	80.98	83.48	83.25	89.51	98.47	96.96	87.76	99.92	95.04	97.34	97.72
Spartina marsh	89.74	97.21	97.20	86.03	92.61	93.14	85.98	83.45	92.90	99.79	99.83	91.30	99.41	97.81	99.14	97.95
Cattail marsh	94.76	96.19	100.0	74.32	99.55	98.84	97.13	94.81	97.31	99.29	100.0	79.38	99.72	99.83	99.50	99.20
Salt marsh	94.71	99.95	99.87	94.86	95.62	98.48	96.53	92.37	94.37	99.92	100.0	97.36	99.53	99.11	99.84	99.79
Mud flats	90.78	92.02	86.58	82.26	93.09	81.38	91.70	89.74	95.56	99.23	95.65	87.59	99.12	91.91	98.43	99.10
Water	100.0	99.99	100.0	97.83	99.54	100.0	99.53	99.93	99.95	100.0	100.0	99.27	100.0	100.0	99.89	100.0
OA	86.06	91.25	92.68	82.01	92.39	89.66	87.25	87.83	91.23	97.33	97.32	87.30	99.15	95.29	96.04	97.55
AA	80.94	87.74	89.63	76.14	91.52	87.68	82.01	83.28	87.60	95.54	95.47	82.78	99.05	93.18	93.37	96.10
Kappa	0.845	0.902	0.918	0.799	0.922	0.885	0.858	0.865	0.902	0.970	0.970	0.859	0.989	0.948	0.956	0.973

with RS, the OA increases to 80.66%. For the MRF classifier, when nonoverlapped sampling is changed to RS, the OA increases from 73.01% to 87.59%, the difference is 14.58%. Compared these results with that of 5% training samples, there is no big difference. This means that even very few samples are used, the RS strategy still can bring optimistic results with the high correlation between training and test samples.

IV. COTRAINING-BASED NONOVERLAPPED HSI CLASSIFICATION METHOD

As shown in Tables I–III, when the nonoverlapped sampling strategy is adopted, the classification performance of HSI dropped dramatically. The main reason is the high spectral correlation between training samples and test samples is removed. When CRS is adopted, if we use the training samples to train the classifier directly, we cannot obtain satisfied results only with the spectral-spatial fusion. Because the spectral feature of training and test samples is not similar enough. So, it is necessary to develop a new classification method for nonoverlapped HSI classification.

We have known that spectral-spatial fusion and spatial adjacency are key to HSI classification. However, with CRS sampling, the spatial adjacency does not exist any longer and the spectral-spatial fusion cannot provide satisfactory classification performance. In this article, we propose an iterated strategy to introduce spatial adjacency during the training process. In this way, spectral-spatial fusion and spatial adjacency can be combined to improve the nonoverlapped HSI classification. The basic idea is that we do not classify the test samples in one step. We can train a classifier with the initial training set, then, the trained model is used to classify the test samples and these classified samples with high confidence are added into the training set. This process can be repeated multiple times to reach a better classification result. With this method, spatial adjacency can be introduced gradually during the training process. However, if only one classifier is used, because the labeled results are highly related to the training samples, the new discrimination information brought by the added training samples is limited and the accumulative error may deteriorate the classification performance.

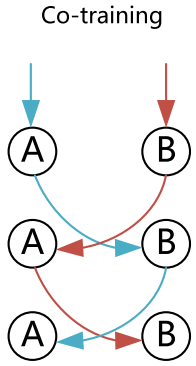


Fig. 7. Cotraining-based nonoverlapped HSI classification.

In this article, in order to improve the nonoverlapped HSI classification performance, two different classifiers are used to introduce diversity and avoid accumulative error. That is, the cotraining strategy is adopted. Specifically, two different classifiers are trained together, then, each trained classifier is used to label the test samples and reliable samples are selected to add into the training set of the other classifier, this process is shown in Fig. 7. In fact, we can use a different classifier with the same feature, or use the same classifier with different features. In this article, EMP and EPF are used as input features and the SVM is used as the classifier. It needs to note that, though the samples with high confidence are added for training, we cannot guarantee that all added samples are with the correct labels. So, in order to reduce the influence of these wrong labeled samples, all samples except these original training samples will be classified with the updated classifier during iteration.

A. Classification Performance of Nonoverlapped Classifier

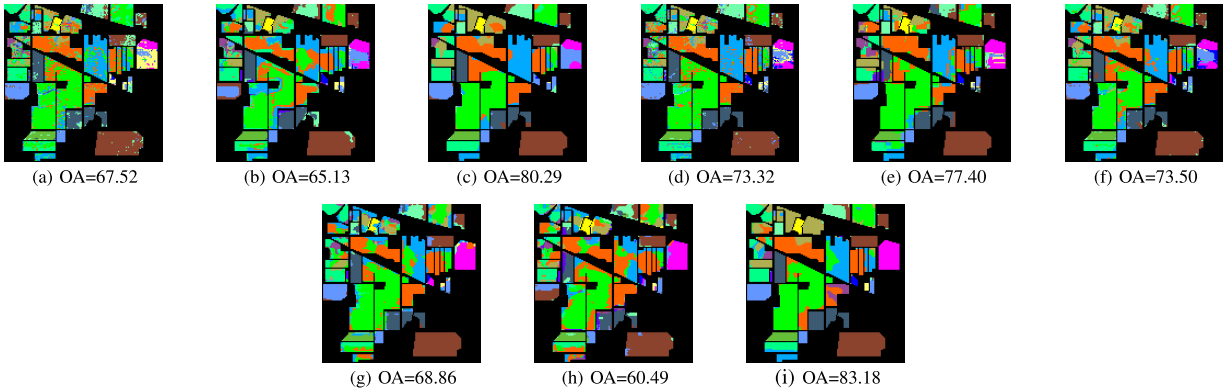
In this section, the Indian Pines, Salinas, and KSC are also used for classification performance evaluation. For Indian

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TABLE IV

SALINAS: THE CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS WITH 0.2% TRAINING SAMPLES

	Non-overlapped Sampling								Random Sampling							
Categories	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN
Weeds 1	98.16	87.75	100.0	99.85	93.91	99.31	85.29	77.56	99.38	96.08	100.0	99.20	98.22	100.0	99.54	92.41
Weeds 2	90.75	89.29	98.01	98.50	96.13	94.24	89.29	86.91	96.89	97.57	99.96	99.59	98.79	98.79	95.34	92.12
Fallow	73.90	57.85	94.92	84.70	83.17	75.34	61.57	60.49	82.36	83.32	95.07	92.79	93.32	85.94	86.89	81.24
Fallow-plow	97.74	91.58	77.55	97.98	91.40	97.60	97.16	76.45	98.09	94.17	68.23	98.38	92.88	98.19	98.75	90.32
Fallow-smooth	83.43	83.49	87.45	84.63	79.02	83.54	77.07	84.23	93.33	87.92	99.98	89.80	84.88	95.43	87.08	93.77
Stubble	99.87	98.52	99.86	95.61	97.42	99.99	98.93	98.82	99.58	99.88	99.90	99.99	96.24	98.44	99.81	97.98
Celery	92.80	90.63	95.72	92.07	97.10	92.75	85.92	84.19	95.47	93.95	96.65	99.89	98.90	95.90	94.69	92.83
Grape	56.19	66.35	53.41	67.55	69.17	55.56	63.87	70.78	65.41	73.38	79.36	84.55	75.25	74.17	72.00	76.14
Soil	79.32	81.81	82.71	95.56	90.10	78.25	88.73	88.83	97.48	95.05	99.39	98.20	95.91	98.32	98.60	95.26
Corn	61.34	68.70	74.57	78.94	91.34	64.09	82.88	71.00	62.70	82.35	65.02	87.83	95.27	67.12	89.68	85.80
Lettuce-4wk	10.00	49.78	0.000	63.44	60.77	45.23	67.02	25.90	93.96	78.89	79.99	90.17	87.98	95.86	87.85	60.23
Lettuce-5wk	81.79	63.00	79.12	77.46	87.90	88.08	84.21	71.96	93.30	82.58	96.10	93.86	98.36	94.14	88.84	89.52
Lettuce-6wk	10.00	80.91	0.000	85.61	73.89	39.46	87.10	65.22	76.53	91.50	20.00	95.46	82.69	98.37	96.71	94.09
Lettuce-7wk	0.000	82.57	0.000	95.87	85.81	19.89	95.44	80.85	86.53	86.61	85.46	98.00	86.08	89.10	97.64	92.65
Vinyard-untrained	39.41	48.69	53.68	62.59	57.55	49.14	43.06	56.56	57.70	65.71	83.35	76.28	69.16	73.32	60.84	74.68
Vinyard-trellis	76.96	95.74	79.71	86.72	99.34	77.78	99.13	72.63	91.40	98.84	95.38	97.55	100.0	91.48	97.83	96.06
OA	69.17	73.17	73.01	79.37	80.00	71.97	74.16	72.51	80.66	84.22	87.59	90.76	87.06	84.55	85.25	85.43
AA	65.73	77.29	67.30	85.44	77.67	72.52	81.67	73.27	86.88	87.99	85.13	93.85	85.57	90.91	90.76	87.82
Kappa	0.653	0.701	0.693	0.769	0.846	0.684	0.711	0.695	0.783	0.824	0.861	0.897	0.909	0.827	0.835	0.838



(a) OA=67.52 (b) OA=65.13 (c) OA=80.29 (d) OA=73.32 (e) OA=77.40 (f) OA=73.50

(g) OA=68.86 (h) OA=60.49 (i) OA=83.18

Fig. 8. Indian pines dataset. Classification maps obtained by (a) SVM, (b) SVM-CK, (c) MRF, (d) EMP, (e) JSRC, (f) EPF, (g) CAE, (h) CRNN, and (i) Cotraining.

Pines and KSC, 10% labeled samples are sampled with CRS for training. For Salinas, 5% labeled samples are sampled with CRS for training. The experimental results are listed in Tables V–VII, respectively. For the Indian Pines, the OA of SVM is only 67.52%. For the classic spectral-spatial methods, the MRF obtains the best performance. The OA is 80.29% and AA is only 70.07%. For the proposed cotraining-based method, the OA is 83.18%, which is only 2.89% higher than that of MRF. However, the AA of cotraining is 84.94%, which is much higher than that of MRF, which is only 70.07%. The accuracy improvement of cotraining benefits from the introduction of spatial adjacency during iteration.

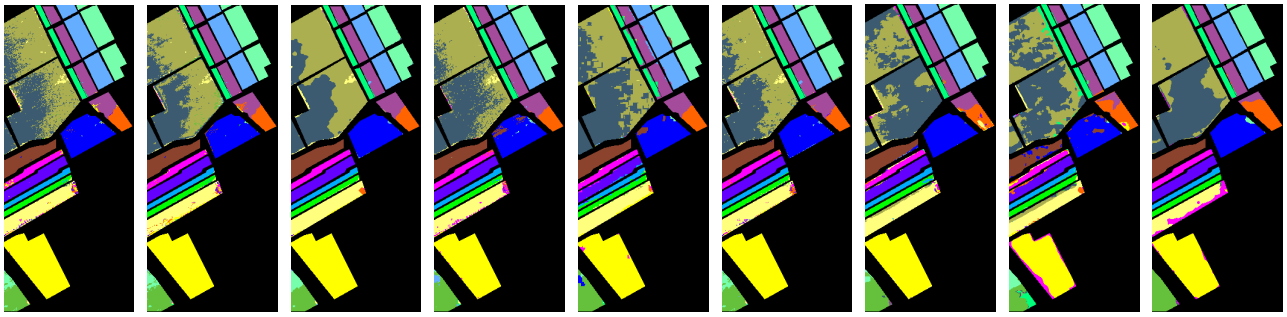
For the Salinas scene, the spatial context still provides consistent performance improvement except the SVM-CK and CRNN. The best performed classic classifier is the EMP, its OA, AA and Kappa are 88.68%,90.25% and 0.874. For comparison, the OA, AA and Kappa of cotraining are 91.60%, 94.70% and 0.907, all of which have an apparent improvement. Similar results can also be found from Table VII. From the above experimental results, we can conclude that cotraining is an effective strategy for nonoverlapped HSI classification by combining spatial adjacency and spectral-spatial fusion.

In order to give a visual presentation of the classification performance of different methods, the classification maps are shown in Figs. 8–10. It can be found that because the spatial information is not used, the classification result of SVM is more scattered compared with other spectral-spatial methods. However, different from RS-based methods, not all the spectral-spatial combinations can improve the classification performance when nonoverlapped sampling is adopted. Such as the SVM-CK, CRNN with Indian Pines, SVM-CK and CRNN with Salinas, EMP with KSC. These results show that some conventional spectral-spatial methods may not suitable for nonoverlapped HSI classification.

B. Iteration Number of Nonoverlapped Classification

For the proposed nonoverlapped classification method, the test samples are gradually included for training. An important parameter is the number of the iteration. In this experiment, the Salinas scene is used to demonstrate the influence

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(a) OA=82.20 (b) OA=81.58 (c) OA=85.45 (d) OA=88.68 (e) OA=84.18 (f) OA=83.93 (g) OA=83.03 (h) OA=79.00 (i) OA=91.60

Fig. 9. Salinas dataset. Classification maps obtained by (a) SVM, (b) SVM-CK, (c) MRF, (d) EMP, (e) JSRC, (f) EPF, (g) CAE, (h) CRNN, and (i) Cotraining.

TABLE V

INDIAN PINES: THE CLASSIFICATION ACCURACY (IN PERCENT) OF THE DIFFERENT METHODS WITH 10% TRAINING SAMPLES

Categories	Non-overlapped Sampling								
	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	Co-training
Alfalfa	43.75	75.10	20.00	59.46	75.64	42.19	73.88	49.36	58.55
Corn-notill	55.47	47.17	68.06	59.85	68.55	61.45	64.85	59.09	68.70
Corn-mintill	57.70	47.34	80.18	57.55	69.00	66.72	59.73	48.08	80.61
Corn	49.63	73.83	94.18	37.91	75.02	77.98	47.34	33.00	49.61
Grass-pasture	74.79	74.98	79.55	70.48	81.38	73.56	71.07	47.22	98.77
Grass-trees	84.44	85.87	95.80	94.09	86.58	91.54	81.27	80.01	99.91
Grass-pasture-mowed	65.87	56.68	27.88	67.86	63.50	76.95	68.57	45.93	93.91
Hay-windrowed	89.31	98.98	96.34	95.16	98.80	95.04	94.19	95.87	100.0
Oats	0.000	43.43	0.000	43.62	13.57	0.000	79.08	19.39	78.03
Soybean-notill	58.88	51.41	74.25	66.29	71.25	64.87	58.31	51.49	86.79
Soybean-mintill	71.55	70.96	86.75	84.25	78.54	80.00	69.74	64.11	91.28
Soybean-clean	49.73	52.41	63.88	50.44	75.79	55.06	46.42	40.37	75.52
Wheat	93.54	98.52	100.0	97.94	93.89	98.57	89.43	78.60	100.0
Woods	88.15	86.35	91.56	93.37	84.85	89.47	83.21	78.07	100.0
Buildings-Grass-Trees	56.73	75.79	84.86	79.72	81.47	69.40	62.05	57.32	78.32
Stone-Steel-Towers	96.47	97.67	57.90	98.33	69.13	98.40	87.49	83.54	99.03
OA	67.52	65.13	80.29	73.32	77.40	73.50	68.86	60.49	**83.18**
AA	64.75	71.03	70.07	72.27	74.16	71.32	71.04	58.22	**84.94**
Kappa	0.630	0.605	0.775	0.697	0.742	0.694	0.643	0.548	**0.811**

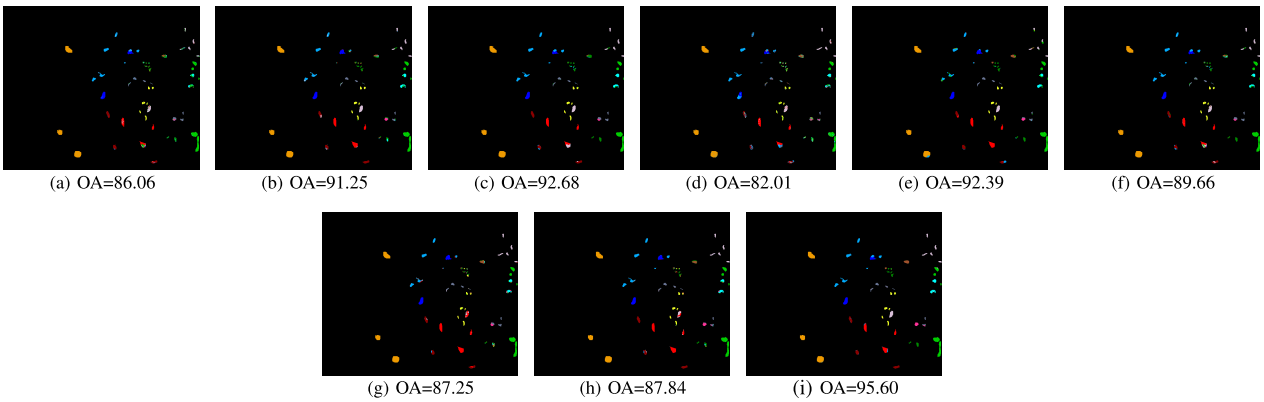
TABLE VI

SALINAS: THE CLASSIFICATION ACCURACY (IN PERCENT) OF THE DIFFERENT METHODS WITH 5% TRAINING SAMPLES

Categories	Non-overlapped Sampling								
	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	Co-training
Weeds 1	96.68	92.55	100.0	97.19	98.37	99.39	94.93	84.16	100.0
Weeds 2	98.44	98.53	99.53	99.24	98.51	98.97	97.95	91.33	99.07
Fallow	90.71	83.69	95.86	95.25	95.66	93.22	88.00	76.48	89.42
Fallow-plow	97.71	96.10	98.15	98.29	92.00	97.92	98.04	87.82	99.37
Fallow-smooth	88.10	78.34	89.56	88.58	72.39	88.87	84.12	90.10	88.88
Stubble	99.95	99.86	99.97	99.83	95.08	99.95	99.03	99.58	99.95
Celery	94.77	92.06	96.03	97.94	98.28	95.34	93.15	87.68	97.81
Grape	70.89	72.67	74.91	82.33	72.81	73.66	75.58	73.13	88.25
Soil	98.30	96.36	99.33	98.97	94.42	99.06	98.61	94.83	99.65
Corn	77.07	87.33	82.17	82.48	93.10	77.37	88.27	86.69	93.76
Lettuce-4wk	83.44	79.88	83.01	77.43	82.63	77.07	88.90	61.06	85.33
Lettuce-5wk	86.47	85.85	93.81	95.83	96.09	90.23	85.68	81.95	99.37
Lettuce-6wk	96.13	88.07	99.21	97.76	77.40	97.38	97.26	88.39	99.39
Lettuce-7wk	94.09	91.57	96.48	96.22	78.97	93.94	98.82	90.12	96.30
Vinyard-untrained	54.67	54.46	64.36	76.79	63.78	61.69	53.82	52.73	78.65
Vinyard-trellis	99.69	96.21	100.0	99.99	100.0	99.48	97.45	90.04	99.96
OA	82.20	81.58	85.45	88.68	84.18	83.93	83.03	79.00	**91.60**
AA	89.19	87.10	92.02	90.25	82.34	90.25	89.98	83.51	**94.70**
Kappa	0.802	0.795	0.838	0.874	0.881	0.821	0.812	0.767	**0.907**

of iteration number. As shown in Fig. 11, with the advance of iteration, the classification accuracy increases gradually. However, when the iteration number increases, the classification performance becomes stable. So, in all the above

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(a) OA=86.06 (b) OA=91.25 (c) OA=92.68 (d) OA=82.01 (e) OA=92.39 (f) OA=89.66

(g) OA=87.25 (h) OA=87.84 (i) OA=95.60

Fig. 10. KSC dataset. Classification maps obtained by (a) SVM, (b) SVM-CK, (c) MRF, (d) EMP, (e) JSRC, (f) EPF, (g) CAE, (h) CRNN, and (i) Cotraining.

TABLE VII

KSC: THE CLASSIFICATION ACCURACY (IN PERCENT) OF THE DIFFERENT METHODS WITH 10% TRAINING SAMPLES

Categories	Non-overlapped Sampling								
	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	Co-training
Scrub	90.76	96.31	98.03	91.94	86.58	95.23	94.28	96.32	99.05
Willow swamp	81.17	86.67	90.72	63.40	96.35	88.81	85.45	84.10	94.08
Cabbage hammock	86.05	92.13	98.52	76.41	91.17	97.25	66.09	78.91	97.50
Cabbage Oak	57.68	60.67	63.37	52.68	77.12	61.70	55.26	60.49	83.36
Slash pine	56.67	62.48	65.59	72.76	91.13	67.41	56.74	65.12	91.52
Broadleaf hammock	57.83	81.42	92.02	62.11	81.94	86.55	73.44	78.18	98.17
Hardwood swamp	75.63	92.38	86.67	51.80	97.23	90.06	80.53	75.99	84.46
Graminoid marsh	76.43	83.18	86.59	83.41	96.47	80.98	83.48	83.25	89.46
Spartina marsh	89.74	97.21	97.20	86.03	92.61	93.14	85.98	83.45	99.20
Cattail marsh	94.76	96.19	100.0	74.32	99.55	98.84	97.13	94.81	99.22
Salt marsh	94.71	99.95	99.87	94.86	95.62	98.48	96.53	92.37	100.0
Mud flats	90.78	92.02	86.58	82.26	93.09	81.38	91.70	89.74	90.04
Water	100.0	99.99	100.0	97.83	99.54	100.0	99.53	99.93	100.0
OA	86.06	91.25	92.68	82.01	92.39	89.66	87.25	87.84	**95.60**
AA	80.94	87.74	89.63	76.14	91.52	87.68	82.01	83.28	**94.31**
Kappa	0.845	0.902	0.918	0.799	0.922	0.885	0.858	0.865	**0.951**

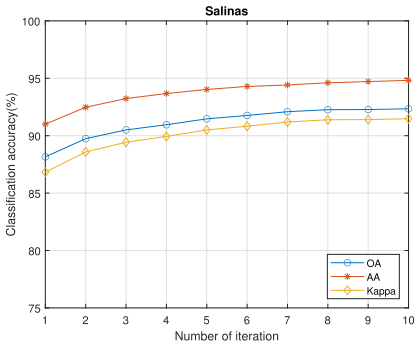


Fig. 11. Influence of iteration number for the nonoverlapped HSI classification.

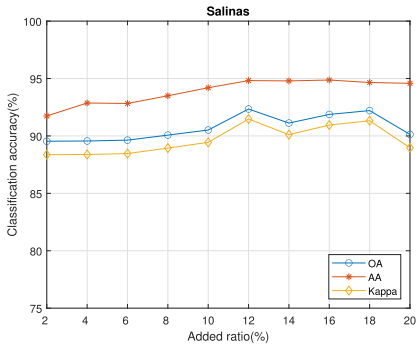


Fig. 12. Influence of added sample number for the nonoverlapped HSI classification.

experiments, the iteration stopped after 10 iterations. This result indicates that by classifying all added training samples during iteration, the accumulative error can be eliminated and avoid performance deterioration.

C. Added Number of Training Samples During Iteration

For cotraining method, the added number of training samples during iteration also has an apparent influence on the nonoverlapped HSI classification. Fig. 12 lists the result of Salinas when different samples are added during the iteration. We can see that the classification accuracy increases with more added samples, this is because more samples can provide more discrimination information for the classifier. However, when too many samples are introduced for training, the reliability of the added samples cannot be guaranteed, so the classification performance may be ruined.

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TABLE VIII
INDIAN PINES: THE CLASSIFICATION ACCURACY (IN PERCENT) OF THE DIFFERENT METHODS WITH STANDARD TRAINING SAMPLES

	Non-overlapped Sampling								
Categories	SVM	SVM-CK	MRF	EMP	JSRC	EPF	CAE	CRNN	Co-training
Alfalfa	52.00	76.00	100.0	84.00	92.00	88.00	71.20	65.20	100.0
Corn-notill	80.30	85.48	92.89	85.63	72.15	89.93	84.00	77.08	93.04
Corn-mintill	63.12	86.88	91.09	89.11	67.82	83.17	82.85	64.38	93.81
Corn	61.62	62.63	66.67	46.46	70.71	60.61	42.63	45.25	100.0
Grass-pasture	85.77	82.12	95.62	83.94	89.78	88.32	76.61	59.49	89.05
Grass-trees	96.61	98.87	99.15	98.87	72.03	98.87	98.64	88.50	100.0
Grass-pasture-mowed	00.00	00.00	00.00	50.00	00.00	00.00	00.00	00.00	00.00
Hay-windrowed	99.20	100.0	100.0	100.0	100.0	99.60	97.12	95.56	100.0
Oats	50.00	70.00	00.00	50.00	00.00	90.00	82.00	68.00	100.0
Soybean-notill	77.73	93.04	93.44	88.67	62.23	90.06	93.54	85.43	97.02
Soybean-mintill	86.95	81.97	92.02	86.10	89.11	89.39	88.76	81.62	94.65
Soybean-clean	76.24	71.99	87.94	76.60	71.63	83.33	71.56	67.27	91.84
Wheat	97.50	97.50	98.75	95.00	50.00	98.75	99.25	97.00	96.25
Woods	87.52	97.98	99.63	99.27	97.80	98.17	94.61	92.31	99.45
Buildings-Grass-Trees	75.76	43.43	90.91	41.41	10.10	75.76	78.79	66.16	98.99
Stone-Steel-Towers	84.09	93.18	75.00	77.27	79.55	77.27	95.45	87.05	90.91
OA	82.78	86.63	93.23	87.29	78.26	89.96	87.12	79.32	**95.54**
AA	73.40	77.57	80.19	78.27	64.06	81.95	78.56	71.27	**90.31**
Kappa	0.803	0.848	0.923	0.855	0.752	0.886	0.853	0.764	**0.949**

D. Experimental Results With Standard Training Samples

To further illustrate the effectiveness of the proposed cotraining method, the Indian Pines with standard training samples are used for comparison. The training samples can be dowloaded from http://dase.grss-ieee.org/. The experimental results are listed in Table VIII. It can be seen that except JSRC and CRNN, all other methods can improve the performance with spectral-spatial fusion. However, the proposed cotraining method still obtains the best performance by appropriate usage of spectral and spatial information of HSI.

V. CONCLUSION

Though RS is still the most used sampling strategy in HSI classification, it is not feasible for practical applications. So, a nonoverlapped sampling strategy—CRS is introduced in this article. With several diverse and representative classification methods, we present the classification results with RS and CRS strategy. Four important factors of HSI classification: spectral feature, spectral-spatial fusion, spatial adjacency, and sample overlap are introduced to analyze the classification results. The spatial adjacency and spectral-spatial fusion are the two most important factors for HSI classification. Then, based on the above analysis, a cotraining-based method is proposed to improve the performance of nonoverlapped HSI classification. We hope this article can provide a feasible solution for practical HSI applications.

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